

Design and Analysis of Algorithms CSEL30503MJ

Module - 1

Background: Review of Algorithm

Content

- What is Design, Analysis & Algorithm?
 - Properties of an Algorithm
 - How to write an Algorithm?
 - Algorithm vs Program
 - Performance Analysis
 - Time Complexity classification
- Best case, Worst case and Average case
- Compute Time complexity to find Sum of individual digits in a number.
- Compute Time complexity to check given number is Palindrome or not.
 - Compute Time complexity for the given algorithms
 - Suggestive Questions

What is Design, Analysis & Algorithm?

- Design: Design an algorithm that minimizes the cost.
- Analysis: Predict the cost of an algorithm in terms of resources and performance.
- Algorithm: Step by step procedure to solve a computational problem is called an Algorithm.

OR

- An Algorithm is a step-by-step plan for a computational procedure that possibly begins with an input and yields an output value in a finite number of steps in order to solve a particular problem.

Properties of an Algorithm

- **TO EVALUATE AN ALGORITHM, WE HAVE TO SATISFY THE FOLLOWING CRITERIA:**

- 1. INPUT:** The Algorithm should be given zero or more input.
- 2. OUTPUT:** At least one quantity is produced.
- 3. DEFINITENESS:** Each instruction is clear and unambiguous.
- 4. FINITENESS:** The algorithm terminates after a finite number of steps.
- 5. EFFECTIVENESS:** Every instruction must be basic i.e. simple instruction.

How to write an Algorithm?

- Algorithm to find the maximum among 3 numbers.

Step1: Start

Step2: Read 3 numbers a, b & c

Step3: If $a > b$ then go to Step4, otherwise go to Step5

Step4: If $a > c$ then print “a is maximum”, otherwise print “c is maximum”

Step5: If $b > c$ then print “b is maximum”, otherwise print “c is maximum”

Step6: Stop

Class Work: WAA to find the minimum among 4 numbers

Algorithm vs Program

Algorithm	Program
1. It is a step-by-step procedure for solving the given problem.	1. A program is nothing but a set of instructions or executable code.
2. An algorithm is designed by Designer	2. The program can be implemented by a programmer.
3. An algorithm is done at design phase	3. A program is implemented at implementation phase.
4. An algorithm can be expressed by using English, flowchart and pseudo code.	4. A program can be written by using languages like C, C++, Java etc....
5. After writing the algorithm, we have to analyze it using space and time complexities.	5. After writing a program, we have to test them

Performance Analysis

The efficiency of an algorithm is declared by measuring the performance of an algorithm. Performance of an algorithm can be computed using Space and Time complexities. Given algorithm can be analyzed in two ways:

1. Space Complexity: The space complexity of an algorithm is the amount of space or memory it needs to run to compilation.
2. Time Complexity: The time complexity of an algorithm is the amount of computer time it needs to run to compilation.

Note: We need to focus on the performance of an algorithm based on time complexity.

Time Complexity classification

- Constant: This statement will be executed by the compiler only once. For example, $c:=a+b$;
- Linear: This statement will be executed by the compiler n number of times.

for $i := 1$ to n step do ----- n times

Statement; ----- n times

- Quadratic: This statement will be executed by the compiler $n*n$ times that is n^2 times.

for $i := 1$ to n step do ----- n times

for $j := 1$ to n step do ----- n times

Statement; ----- n^2 times

Time Complexity classification

- Cubic: This statement will be executed by the compiler $n*n*n$ times that is n^3 times.

for i := 1 to n step do ----- n times

for j := 1 to n step do -----n times

for k := 1 to n step do-----n times

Statement; ----- n^3 times

- Logarithmic: For each and every time the work area will be sliced to half. In such cases time complexity will be $\log n$.

Best case, Worst case and Average case.

- If an algorithm takes minimum amount of time to complete for a set of specific inputs it is the Best case. For example, 'key' element is found at beginning of an array in linear search.
- If an algorithm takes maximum amount of time to complete for a specific set of inputs it is worst case. For example, 'key' element is found at end of an array or element not found.
- If an algorithm takes average amount of time to complete for set of specific inputs it is average case. For example, 'key' element found at middle of an array in linear search.

Compute Time complexity to find Sum of individual digits in a number.

Statement	Time Complexity
Algorithm Sumofindiviual(n) {	- -
While n > 0 do {	m -
k:= n % 10;	m
n:= n / 10;	m
s:= s + k; }	m -
Return s; }	1 -

TOTAL COMPLEXITY: $m+m+m+m+1=4m+1=O(m)$

Compute Time complexity to check given number is Palindrome or not.

Statement	Time Complexity
Algorithm Palindrome(n)	-
{	-
m:= n;	1
While n > 0 do	m
{	-
k:= n % 10;	m
n:= n / 10;	m
s:= (s*10) + k;	m
}	-
If s = m then	1
Print "given number is Palindrome"	1
Else	0
Print "given number is not Palindrome"	0

Compute Time complexity for the following algorithms

- Compute Time complexity to check given number is Armstrong or not.
- Compute Time complexity to check given number is prime or not.
- Compute Time complexity to find Fibonacci sequence up to given number.
- Compute Time complexity to find GCD of two numbers.
- Compute Time complexity to find factorial of a given number.
- Compute Time complexity to find sum of elements present in an array.
- Compute Time complexity to perform Linear Search.

Suggestive Questions

- Define an algorithm. What are the different criteria that satisfy the algorithm?
- Explain how algorithms performance is analyzed?
- What are the different techniques to represent an algorithm? Explain.
- Distinguish between Algorithm and Program.
- Write an algorithm to find the sum of individual digits of a given number
- What is meant by time complexity? What is its need? Explain different time complexity notations. Give examples one for each.
- Describe the performance analysis in detail
- What is meant by space complexity?